

The PGR Domain

A developer’s guide to what the platform actually models — how a research degree works in the real world, who the people are, and why every screen exists.

Read this before clicking through the app. Every feature in the platform is a digitisation of a process that already happens at a university — usually in spreadsheets, email and institutional memory. Once you know the process, the screens explain themselves: when you press “Decide” you are playing a progression panel; when a second person confirms an award you are playing the exam board. The platform follows the UK model (“PGR” = Postgraduate Research: PhD and research master’s students), which is also the model HESA reporting assumes.

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CHAPTERS

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KINDS OF ACTOR

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YEARS MODELLED

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PERSON RECORD THROUGHOUT

A research degree involves five kinds of actor. Every permission, screen and separation-of-duties rule in the platform maps to one of them.

ACTOR	IN THE REAL WORLD	IN THE PLATFORM
PGR office / Registry	Administrators who run the machinery: admissions paperwork, student records, deadlines, regulatory returns, committee logistics. Registry is the record-keeping heart of a university; the PGR office is its research-degrees arm.	The <code>PGR Administrator</code> role (day-to-day operations) and <code>Institution Administrator</code> (plus configuration). Statutory sign-off deliberately sits with Registry-type permissions, not general admin.
Supervisors	Academics who guide the research. Every student has a <i>primary</i> supervisor and usually a <i>co-supervisor</i> — so one person's absence, blind spot or bias cannot sink a student. Supervisors also sit on progression panels.	The <code>Supervisor</code> role: sees only their own caseload (row-scoped), logs meetings, holds <code>progression.decide</code> .
Panels & boards	Small committees for the high-stakes judgements: progression panels, examination panels, the exam board that confirms awards. Deliberately <i>not</i> the supervisor alone — independence is the point.	Panel membership recorded per milestone; approver-separation rules (proposer ≠ confirmer, trainer ≠ approver) are the digital form of this.
Funders	Research councils (UKRI), charities, industry. They grant the university an award — a pot of money with dates — which pays student stipends (a tax-free living allowance, roughly £18–20k/yr) and sometimes fees.	Funders and awards are <i>references</i> : the platform holds the relationships and dates, finance systems hold the actual money. That honesty is deliberate.
HESA	The national statistics agency. Every UK university must submit an annual coded return about every student. Errors are a compliance problem with real consequences.	The Statutory screen: field mappings held as data, a readiness gate, and sign-off as an attestation act.

Why one person record matters: the same human being appears over the years as applicant, student, teaching assistant (employee), alumnus — and later maybe examiner or supervisor. Most universities hold each of those in a different system and reconcile forever. The platform's core design decision is one `person_id` carrying dated identities, so the thread never breaks.

STAGE 1 **Demand and money come first.** A department doesn't admit PhD students at random: someone says "we need a researcher on X" (a **research demand**), and there is usually a funding **award** to pay for it. Only then is a **position advertised** (an "opportunity").

Why the app links opportunity → award on day one: three years later the funder will ask "which students did our £2M pay for?". In most institutions the answer is reconstructed from email. Here the lineage starts at the advert, so the question is a query.

STAGE 2 **Application and assessment.** Candidates apply (each becoming a person record, created once). The office shortlists, gathers **references** (external referees answer via tokened links — no accounts), runs **interviews** with panellists, and records an **assessment** with a rationale. Every stage advance carries a reason, because rejections get challenged.

Offer and conditions. A successful candidate gets an **offer**, almost always conditional: "subject to your master's certificate, your English test, your visa". Each condition is tracked to met/waived. The admissions office's recurring nightmare — "who still has unmet conditions two weeks before term?" — becomes a filter instead of an inbox search.

The identity moment. When the offer is accepted, the student record is created *on the applicant's existing person record*, in one transaction. In a two-system estate this is exactly where identity breaks and every downstream report starts disagreeing.

In the app: Research (demands, awards) → Recruitment (opportunities, pipeline, application detail with Advance/Assess/Offer) → Admissions (queue). Accepting an offer requires `student.write` — an admissions act, not a recruitment act — and fires the opening workflow tasks.

STAGE 3 **Registration** gives the student a start date, a programme, a study mode (full/part-time) and — crucially — an **expected end date**: start + 3–4 years, stretched by the part-time factor. This date is the spine of the whole record:

- funding must cover it (the integrity engine checks exactly this),
- progression reviews are scheduled from the start date,
- “overrunning” it is the completion-rate metric institutions are judged on.

Why the baseline is immutable: the platform stores the *originally agreed* end date forever and only ever *recomputes* the current one as baseline + approved suspensions + extensions (rescaled on mode changes). So the end date can always be explained, and approving the same suspension twice cannot corrupt it. Nobody ever types an end date by hand.

Interruptions are normal, not exceptional. Over four years, life happens: **suspensions** (illness, parental leave — the clock stops), **extensions** (more time at the end), **mode changes** (full↔part time, which rescales the remaining duration). Each is *requested* by one person and *approved* by another — a welfare-adjacent decision nobody should make alone — and each shifts the computed end date.

In the app: the Record section + Lifecycle changes panel on the student record (`student.lifecycle.approve` is a separate permission from ordinary editing); the Journey tracker at the top renders the whole clock — including “paused since...” — live.

STAGE 4 For three to four years, three loops run side by side. Most of the platform’s operational value lives here.

LOOP 1 — SUPERVISION

The student meets their supervisors regularly; institutions set a cadence policy (e.g. at least every 90 days). Meetings are logged with notes and agreed actions, and the *student confirms* the note — a two-sided record. When a student later fails or complains, “was this student actually supervised?” becomes a quasi-legal question, and the log is the answer. Just as important: **silence is the earliest warning signal** — a student nobody has met for five months is the one to worry about, and the caseload flags exactly that.

LOOP 2 — PROGRESSION REVIEWS (MILESTONES)

A PhD is not continuously examined; it has formal checkpoints. Typical pattern: an **induction review** (~1 month in — is the plan sound?), a **confirmation review** (~9–12 months — historically the “upgrade from MPhil to PhD”: is this genuinely doctoral work?), then **annual reviews**. For each: the student *submits* evidence, a **panel** (not the supervisor alone) decides *progress* / *progress-with-conditions* / *no-progress* with a written rationale and an outcome letter, conditions get signed off, and there is an **appeal** route because the decision can end a career. This is the university’s quality-assurance obligation — the QAA expects it to demonstrably happen.

LOOP 3 — MONEY

The student’s funding is one or more **arrangements** (research-council stipend, university scholarship, external sponsor, self-funded), each with dates, amounts and a contribution percentage, paid in **instalments** (scheduled → approved → paid, with a finance reference), sometimes with **fee waivers**. The classic institutional failure: the arrangement and the registration drift apart — funding ends in March, studies end in October, and nobody notices until the student’s bank account does.

Why nine named rules beat a dashboard: the integrity engine checks the whole chain (funder → award → arrangement → student → dates) and returns findings like `funding_gap: 124 days between 2026-06-07 and 2026-10-09`. A finding an administrator can forward verbatim to a supervisor gets acted on; a score does not.

In the app: Supervision (caseload + meeting log), Progression (programme rules) + the Milestones panel per student, Funding + Funding integrity. Pattern Lab sits beside these loops as the advisory layer: “which current students resemble past students who ran into trouble?” — association, never decision.

STAGE 5

Thesis. The student declares an **intention to submit** (which starts the search for examiners) and then **submits** the thesis — a document reference with a date, because the examination clock runs from evidence, not memory.

Examination. Two **examiners** are appointed: one internal, one **external** from another university — with declared conflicts of interest, because independence must be recorded before the exam, not asserted after. Then the **viva voce**: an oral defence, typically two to four hours of questioning, sometimes with an independent chair. Outcomes are graded: *pass*, *pass with minor corrections* (most common), *major corrections*, *revise and resubmit*, *fail*. Corrections carry deadlines that are tracked and approved.

The award. Bureaucratic finality with separation of powers: a chair **proposes** the classification (“Doctor of Philosophy”), the exam board **confirms** it — the platform enforces that the confirmer is a different person — and Registry **publishes** it. Only a published award can graduate.

Graduation. One act closes every thread consistently: the studentship completes, funding arrangements end, and the person’s student identity closes while the *person* becomes **alumni**. The record lives on — that name will reappear as an examiner, a collaborator, a donor.

In the app: the Thesis & completion panel (intention, submit, examiners, viva, outcome, corrections), the Award classification card (propose → confirm → publish), and Graduate. The Journey tracker turns fully green with a 🎓 badge.

MEANWHILE, ALWAYS IN THE BACKGROUND

- **HESA / statutory returns** — the annual coded return per student; the Statutory screen keeps mappings as data with a mandatory-field gate before sign-off, so a spec change is a remap, not a release.
- **Tasks & workflow** — every event above raises human to-dos with SLAs; the queue is the audit of what humans did.
- **Integration** — finance/HR/research systems hear about changes through a transactional outbox; reconciliation shows pending/dispatched/dead-lettered honestly.
- **Audit** — every write, logged by middleware at the edge, so no developer can forget it.

TERM	MEANING
PGR	Postgraduate Research — PhD and research master's (MPhil, MRes) students, as opposed to taught degrees.
Stipend	The tax-free living allowance a funded student receives (~£18–20k/yr), paid in instalments.
Award (funding)	A grant the university holds from a funder — a pot with a reference, value and dates. Not the degree.
Award (degree)	The classification conferred at the end (PhD, MPhil...). Context disambiguates; the platform models both.
UKRI	UK Research and Innovation — the umbrella for the national research councils, the biggest PGR funders.
Confirmation review	The make-or-break checkpoint ~1 year in; historically the "MPhil→PhD upgrade".
Viva (voce)	The oral defence of the thesis before examiners. Latin for "by live voice".
External examiner	An examiner from another institution — the independence guarantee of UK examination.
Corrections	Post-viva revisions (minor/major) with deadlines; the most common outcome.
Suspension / interruption	An approved pause of study — the clock stops, the expected end moves out.
QAA	Quality Assurance Agency — expects institutions to demonstrate structured progression monitoring.
HESA	Higher Education Statistics Agency — the mandatory annual coded return about every student.
Registry	The institution's official record-keeping function; owns statutory attestation.
Row scoping	Platform term: a supervisor's queries are filtered to their own students server-side; out-of-scope reads return 404, never 403, so existence is not leaked.
Bitemporal	Platform term: funding rows carry valid_from/valid_to; a change closes one row and opens another, so history is never overwritten.
Outbox	Platform term: integration events written in the same transaction as the change, dispatched at-least-once to partners.

When you run the end-to-end walkthrough, this is who you are pretending to be at each step.

REAL-WORLD ACT	WHO REALLY DOES IT	WHERE IN THE APP	YOU CLICK
Student hands in review evidence	The student	Milestones panel	Submit
Progression panel decides	Panel (supervisor + independents)	Milestones panel	outcome progress → Decide
Student declares they will submit	Student + supervisor	Thesis & completion	Declare intention
Thesis handed in	Student	Thesis & completion	Submit thesis
Viva happens, examiners decide	Examination panel	Thesis & completion	outcome pass → Record outcome
Office verifies requirements met	PGR office	Thesis & completion	Confirm completion
Chair proposes the degree	Exam board chair	Award classification	PhD → Propose (as admin)
Exam board confirms	A different board member	Award classification	Confirm (as the student's supervisor — Rosa/Elena)
Registry publishes the award	Registry	Award classification	Publish (as admin)
Graduation ceremony & close-out	Registry	Thesis & completion	Graduate

Two things the platform will refuse — on purpose: confirming an award with the same login that proposed it (approver separation), and deciding “progress with conditions” without written conditions. When you hit these refusals, that is the domain's governance showing through, not a bug.

Watch the side effects: after Graduate, check the Journey tracker (all green, 🟢), the Funding panel (arrangements ended), Persons (student identity closed, person remains — alumni), and /audit (every click you made, attributed to the login that made it).